MATH 111 Calculus 1, NUU, FALL 2026

Dr. Salnikova Chekeres **Quiz 2 · Continuity and limits at infinity (1.8, 3.4)**

12

15 minutes. 10 points. No notes, no calculators, no phones. Write your answers in the spaces provided and show all your work. Write "DNE" if a limit does not exist.

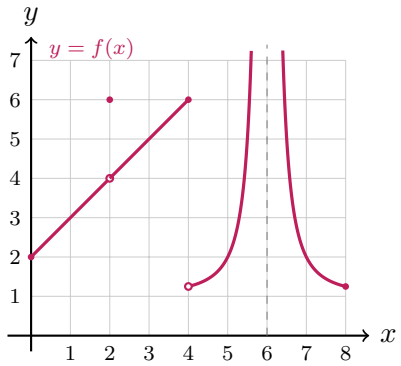
Name: _____

ID: _____

1	2	3	Total

Problem 1 *Reading a graph.* **[4 points]**

The graph of a function f is shown; the dashed line is a vertical asymptote. Answer each of the following.



(a) $\lim_{x\to 2} f(x) =$ _____

(b) $f(2) =$ _____

(c) The numbers at which f is discontinuous: _____

(d) At which of them is the discontinuity *removable*, and what value of f there would remove it?

_____

Problem 2 *From the lectures.* **[2 points]**

(Question 12 of Lecture 6, word for word.) Circle one letter, and show the calculation that gives it.

$\lim_{x\to\infty}\left(\sqrt{x^2+x}-x\right) = ?$

A. 0 — for large x, $\sqrt{x^2+x}$ is practically x.

B. $\frac{1}{2}$

C. ∞ — both terms grow without bound.

D. It does not exist: $\infty - \infty$ is undefined.

Calculation:

Problem 3 *An existence argument.* **[4 points]**

Show that the equation $x^3 - 3x^2 - 6 = 0$ has a solution in the interval $(3, 4)$.

Write the argument in full: name the function, say why it is continuous, give the two values, and state the conclusion.